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Third SEMESTER–B.TECH- Course Work
MID-SEMESTER EXAMINATION, September 2026

Course Code-ICMTC301/EEMTC301/CMMTC301
Course Title-Engineering Mathematics
Time: 1:30 hours Max Marks-20

Note: Attempt all questions. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1 (a)	Express $f(x) = \frac{x}{2}$ as a Fourier series in the interval $-\pi < x < \pi$.	2	CO1
(b)	Find the half range sine series for $f(x) = e^{ax} - e^{-ax}$ in $0 < x < \pi$.	2	CO1
2 (a)	If $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx)$, where $f(x) = \begin{cases} \pi x & 0 \le x \le 1 \\ \pi(2-x) & 1 \le x \le 2. \end{cases}$ Find b_n.	2	CO1
(b)	Analyse harmonically the data given below and find a_0 and a_1,	2	CO1
3 (a)	Find directional derivative of the function $\phi = 4xz^3 - 3x^2yz^2$ at (2,-1,2) along z-axis.	2	CO2
(b)	Determine the constant a so that the vector $\vec{F} = (x+3y)\hat{i} + (y-2z)\hat{j} + (x+az)\hat{k}$ is solenoidal.	2	CO2
4 (a)	Evaluate $\int_C [(2x^2y + y + z^2)\hat{i} + 2(1+yz^3)\hat{j} + (2z + 3y^2z^2)\hat{k}].d\vec{r}$, along the curve $y^2 + z^2 = a^2$, $x = 0$.	2	CO2

x	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π
y	1.98	2.15	2.77	-0.22	-0.31	1.43	1.98

Q. No.	Question	Marks	CO
(b)	Determine whether $\vec{F} = (y^2 \cos x + z^3)\hat{i}$ $+(2y \sin x - 4)\hat{j} + (3xz^2 + 2)\hat{k}$, is a conservative vector field.	2	CO2
5 (a)	Evaluate by Green's theorem, $\int_C e^{-x}(\sin y\ dx + \cos y\ dy)$ where C is the rectangle with vertices (0,0), $(\pi, \frac{\pi}{2})$, $(0, \frac{\pi}{2})$ and $(\pi, 0)$.	2	CO2
(b)	Evaluate $\iint_S \vec{A}.\hat{n}dS$, where $\vec{A} = z\hat{i} + x\hat{j} - 3y^2z\hat{k}$ and S is the surface of the cylinder $x^2 + y^2 = 16$ included in the first octant between $z = 0$ and $z = 5$.	2	CO2

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